Smart Campus Management System

RAG Test Document

1. Introduction

The Smart Campus Management System (SCMS) is a software platform designed to improve the management of university facilities, classrooms, energy consumption, and student services. The system combines sensors, web applications, databases, and analytics to provide administrators with real-time information about campus resources.

The primary goal of SCMS is to reduce operational costs while improving the efficiency and availability of campus facilities. The system is designed for universities with multiple buildings, classrooms, laboratories, libraries, and administrative offices.

SCMS consists of four major components: Sensor Network, Central Management Server, Analytics Engine, and User Dashboard.

2. Sensor Network

The Sensor Network collects information from physical devices installed throughout the campus. The sensors can monitor room temperature, humidity, occupancy, electricity consumption, lighting status, and air quality.

Each sensor periodically sends measurements to the Central Management Server. The default measurement interval is five minutes.

For example, an occupancy sensor may report that a classroom contains 42 people. The server stores this information along with the building name, room number, sensor identifier, and timestamp.

Sensors communicate with the server using a lightweight messaging protocol. The system is designed to continue collecting data even if one sensor temporarily becomes unavailable.

3. Central Management Server

The Central Management Server is responsible for receiving, validating, storing, and distributing sensor data.

When a measurement arrives, the server performs three checks: it verifies that the sensor is registered, checks whether the measurement contains a valid timestamp, and checks whether the reported value falls within an acceptable range.

Invalid measurements are rejected and recorded in an error log. Valid measurements are stored in the campus database. The database contains separate tables for sensors, buildings, rooms, measurements, users, and maintenance records.

The server also provides an API that allows authorized applications to retrieve campus information.

4. Analytics Engine

The Analytics Engine processes historical and real-time data collected by the system.

One primary function is energy analysis. It compares electricity consumption between buildings and identifies unusual increases in energy usage.

For example, if Building A normally consumes approximately 800 kilowatt-hours per day but suddenly consumes 1,250 kilowatt-hours, the Analytics Engine can flag the building for investigation.

The Analytics Engine also calculates classroom utilization using: Utilization = Occupied Hours / Available Hours × 100.

If a classroom is available for 10 hours and occupied for 6 hours, its utilization is 60 percent.

Utilization is classified as Low below 40 percent, Normal from 40 percent to 75 percent, and High above 75 percent.

5. User Dashboard

The User Dashboard provides administrators with a visual representation of campus information. It contains five main views: Campus Overview, Energy Monitoring, Room Occupancy, Sensor Health, and Maintenance.

Campus Overview displays the total number of buildings, active sensors, occupied rooms, and current energy consumption.

Energy Monitoring displays electricity consumption for individual buildings and historical trends. Room Occupancy shows currently occupied classrooms and laboratories and allows filtering by building and occupancy level.

Sensor Health classifies sensors as Online, Offline, or Warning. A sensor is marked Offline if the system has not received a measurement from it for more than 15 minutes.

The Maintenance view contains records of equipment problems and maintenance activities.

6. User Roles

SCMS supports three primary user roles: Administrator, Facility Manager, and Technician.

Administrators have full access to the system. They can view analytics, manage users, register sensors, and create maintenance requests.

Facility Managers can view building information, energy consumption, occupancy information, and maintenance records. They cannot create or delete system users.

Technicians primarily use the system to manage maintenance requests. They can update the status of maintenance tasks but cannot modify sensor registration information.

7. Maintenance Process

When a technical problem is detected, a maintenance request is created. Each request contains a Request ID, Building, Room, Equipment, Description, Priority, Assigned Technician, Creation Date, and Current Status.

Priority levels are Low, Medium, and High. The maintenance workflow contains four statuses: Open, Assigned, In Progress, and Completed.

A request begins with Open. After a technician is assigned, it changes to Assigned. When work begins, it becomes In Progress. After the problem is resolved, the technician marks it Completed.

High-priority requests should normally be investigated within four hours.

8. Security

SCMS uses authentication and role-based authorization. Users must authenticate before accessing the dashboard. The system determines permitted features based on the user's role.

The system records important security events in an audit log. These include successful logins, failed login attempts, changes to user permissions, sensor registration changes, and maintenance record modifications.

9. Data Retention

Sensor measurements are stored in the database for two years. After two years, detailed sensor measurements are moved to archival storage. Summary statistics remain available in the main database so administrators can continue viewing long-term trends.

The archival process runs once every month. Maintenance records are retained for five years because they may be required for equipment history and operational analysis.

10. Example Scenario

Consider a university with three buildings: Engineering Block, Science Block, and Library Block.

The Engineering Block contains 80 sensors, the Science Block contains 65 sensors, and the Library Block contains 35 sensors.

The Engineering Block consumes an average of 1,100 kilowatt-hours per day. The Science Block consumes approximately 850 kilowatt-hours per day, while the Library Block consumes approximately 500 kilowatt-hours per day.

One Monday, the Analytics Engine detects that the Engineering Block consumed 1,750 kilowatt-hours. This is significantly higher than its normal daily consumption. The system flags the Engineering Block for investigation.

A Facility Manager examines the Energy Monitoring dashboard and discovers that electricity consumption increased primarily between 6:00 PM and 10:00 PM.

A maintenance request is created with High priority because an electrical equipment problem is suspected. A technician is assigned to the request. After inspection, the technician discovers that a large laboratory cooling system was accidentally left running after operating hours.

The technician switches off the cooling system and marks the maintenance request as Completed. The following day, the Engineering Block's energy consumption returns to approximately 1,080 kilowatt-hours.

11. System Limitations

SCMS depends on the availability and accuracy of sensor data. A faulty sensor can produce incorrect measurements, which may affect analytics results.

The system can identify unusual energy consumption, but it cannot automatically determine the physical cause of every abnormal event.

Internet connectivity is required for remote access to the dashboard.

The system currently does not provide automatic control of electrical equipment. It only monitors equipment-related information and allows administrators and technicians to record maintenance activities.

12. Important Facts for Testing

Sensor measurements are normally collected every five minutes. A sensor becomes Offline after more than 15 minutes without a measurement. Classroom utilization below 40 percent is Low, 40 to 75 percent is Normal, and above 75 percent is High.

Sensor data is retained in detailed form for two years. Maintenance records are retained for five years. The archival process runs once per month. High-priority maintenance requests should normally be investigated within four hours.

SCMS has three primary user roles and four major technical components. Engineering Block has 80 sensors, Science Block has 65, and Library Block has 35. Engineering Block normally consumes approximately 1,100 kWh per day and consumed 1,750 kWh during the abnormal Monday. The suspected cause was a laboratory cooling system left running after operating hours. SCMS does not currently provide automatic control of electrical equipment.

13. Information Not Provided

The document does not specify the programming language used to develop SCMS, the name of the database management system, the exact messaging protocol, the total number of students, the cost of installing the sensor network, the manufacturer of the sensors, the name or location of the university, the technician's name, or the exact hardware specifications of the Central Management Server.